

Concrete Strength Prediction: Final Report

DATA 4950 Data Science Capstone — Spring 2026

Team 6

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1. Executive Summary

Concrete compressive strength is a key performance measure in concrete mix design because it indicates whether a mixture can safely support expected structural loads. This project predicted concrete compressive strength, measured in megapascals (MPa), from concrete mixture ingredients and curing age. The goal was to support early-stage screening of candidate concrete mixes before laboratory validation. A model that overestimated strength could encourage unsafe mix decisions, while a model that underestimated strength could lead to over-design, higher material cost, and unnecessary cement use.

The team used the UCI Concrete Compressive Strength dataset originally associated with Yeh (1998). The raw dataset contained 1,030 concrete mix records and 9 columns: 8 predictors describing mixture components and curing age, plus the compressive strength target. The predictors represented cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age. After preprocessing removed 25 duplicate rows, the final modeling dataset contained 1,005 rows. The data were split into 804 training observations and 201 test observations using an 80/20 split.

The best model was a tuned Gradient Boosting regression model. It achieved a test R^2 of 0.9331 and a test RMSE of 4.0616 MPa. Compared with the naive training-mean baseline RMSE of 15.71 MPa, the tuned model reduced error by about 11.64 MPa. The model also performed consistently across five-fold cross-validation, with a mean R^2 of 0.9230 and a standard deviation of 0.0192. These results indicated that the final model was not simply the result of a favorable train-test split.

The most actionable finding was that concrete strength depended most strongly on the interaction between cementitious binder content and curing age. The top feature in the final model was the binder-age interaction, with feature importance 0.5234. The second and third most important features were water-to-binder ratio, with importance 0.1663, and cement-age interaction, with importance 0.1472. In practical terms, the model suggested that candidate mixes should be screened by evaluating the balance among binder content, curing time, and water relative to binder, rather than by evaluating ingredient amounts independently.

2. Problem Statement and Data Description

The prediction task was to estimate the compressive strength of a concrete mix from known mixture proportions and curing age. The intended user was a concrete producer, project engineer, or analyst comparing candidate mix designs before conducting full laboratory or field validation. The real-world consequence of error was asymmetric and meaningful.

Overestimating strength could create a safety risk if a mix were believed to meet performance requirements when it did not. Underestimating strength could lead to unnecessary redesign, excess cement content, higher cost, and avoidable environmental impact.

The dataset came from the UCI Concrete Compressive Strength dataset. It contained observations for concrete mixes with 8 numeric input variables and one numeric target. The input variables measured cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and curing age. The target variable measured concrete compressive strength in MPa. The raw dataset contained 1,030 rows and 9 columns. After duplicate removal, 1,005 rows remained for modeling.

The target variable ranged from 2.33 MPa to 82.60 MPa in the raw EDA output. Its mean was 35.82 MPa and its median was about 34.45 MPa, indicating a mildly right-skewed distribution. After duplicate removal, the cleaned dataset had a target mean of 35.25 MPa, median of 33.80 MPa, and the same overall target range of 2.33 MPa to 82.60 MPa. The broad range showed that the dataset included very low-strength and high-strength concrete mixes, which made regression error interpretation important in practical units rather than only through R^2 .

No data-leakage columns were dropped. All predictors described mixture components or curing age that would be known before or during the compressive strength testing process. The target column itself was separated from the predictors before modeling. Because the dataset contained only numeric mixture and curing variables, no identifier, post-outcome measurement, or target-derived column required removal for leakage prevention.

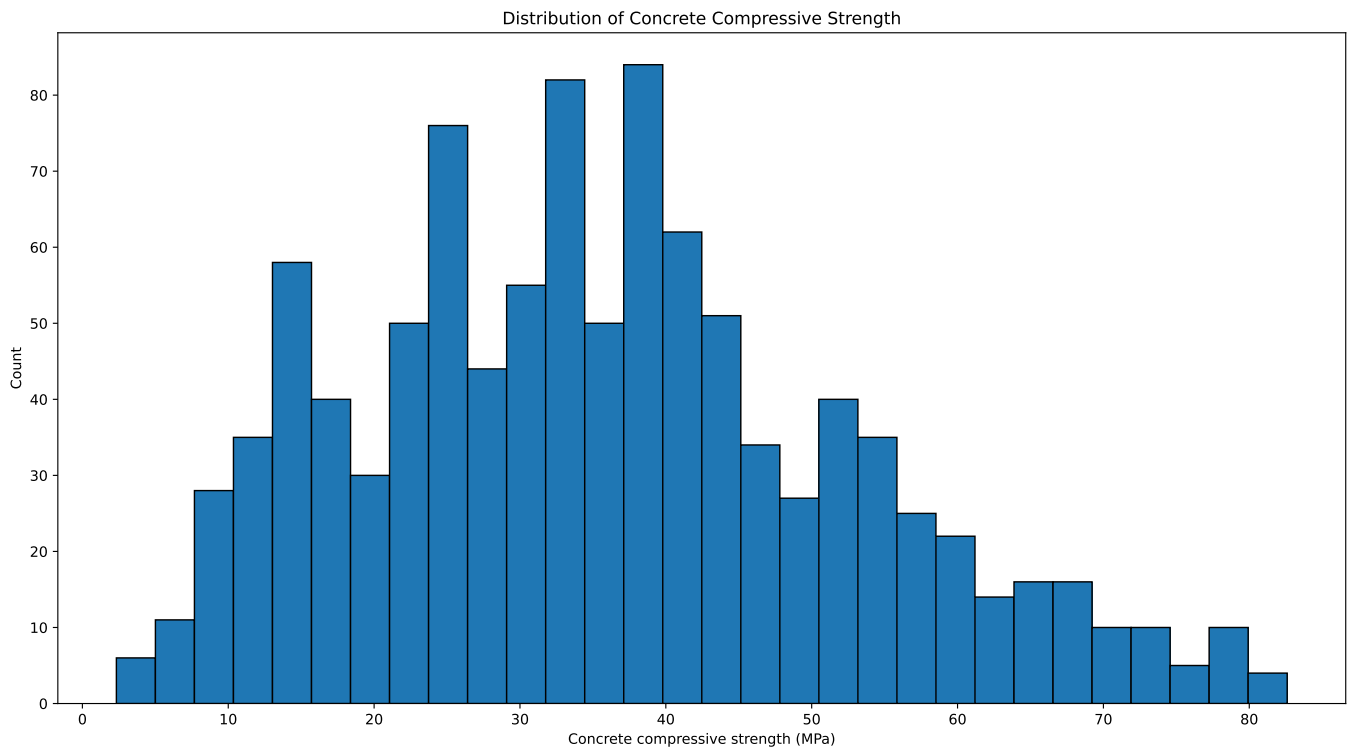


Figure 1. Distribution of concrete compressive strength in MPa. The target ranged from 2.33 to 82.60 MPa and was mildly right-skewed, with a mean of 35.82 MPa and median near 34.45 MPa in the raw EDA output.

3. Exploratory Data Analysis

The first important EDA finding was that the target variable had a wide but realistic strength range. Compressive strength ranged from 2.33 MPa to 82.60 MPa. The raw target mean was 35.82 MPa and the median was about 34.45 MPa, so the distribution was mildly right-skewed. This mattered because the project needed an error metric that could be interpreted in MPa. RMSE was therefore useful because it measured prediction error in the same unit as the target.

The second finding was that several predictors were strongly skewed or contained many zero values. Blast furnace slag, fly ash, superplasticizer, and age were right-skewed. Some mixes contained no slag, fly ash, or superplasticizer, while other mixes used substantial amounts. Age also had a long upper tail, with values extending to 365 days even though the median curing age in the cleaned data was 28 days. This pattern later supported the use of nonlinear transformations and interaction features involving age.

The third finding was that the predictors were measured on very different scales. Coarse aggregate averaged roughly 974 kg/m³ after duplicate removal, while superplasticizer averaged only about 6.03 kg/m³. This scale difference did not directly harm tree-based models, but it mattered for linear models and regularized regression. The team therefore preserved both unscaled and scaled versions of the engineered feature set. Scaled features were used for linear and regularized models, while unscaled features were available for tree models.

The fourth finding was that the strongest raw linear relationship with compressive strength came from cement. Cement had a correlation of approximately $r = 0.50$ with strength in the EDA. Superplasticizer and age also had positive correlations, approximately $r = 0.37$ and $r = 0.33$, respectively. Water had the strongest negative raw relationship with strength, approximately $r = -0.29$. These relationships matched domain expectations: more cementitious material and longer curing generally increased strength, while excess water tended to weaken concrete.

The fifth finding was that some predictors were correlated with each other, which created possible multicollinearity for linear models. Water and superplasticizer had a relatively strong negative correlation of approximately $r = -0.66$. The feature engineering notebook later identified additional high-correlation relationships among engineered predictors, especially among inverse or complementary ratio features. This supported the conservative pruning step used after feature creation because the final model did not need multiple columns carrying nearly the same information.

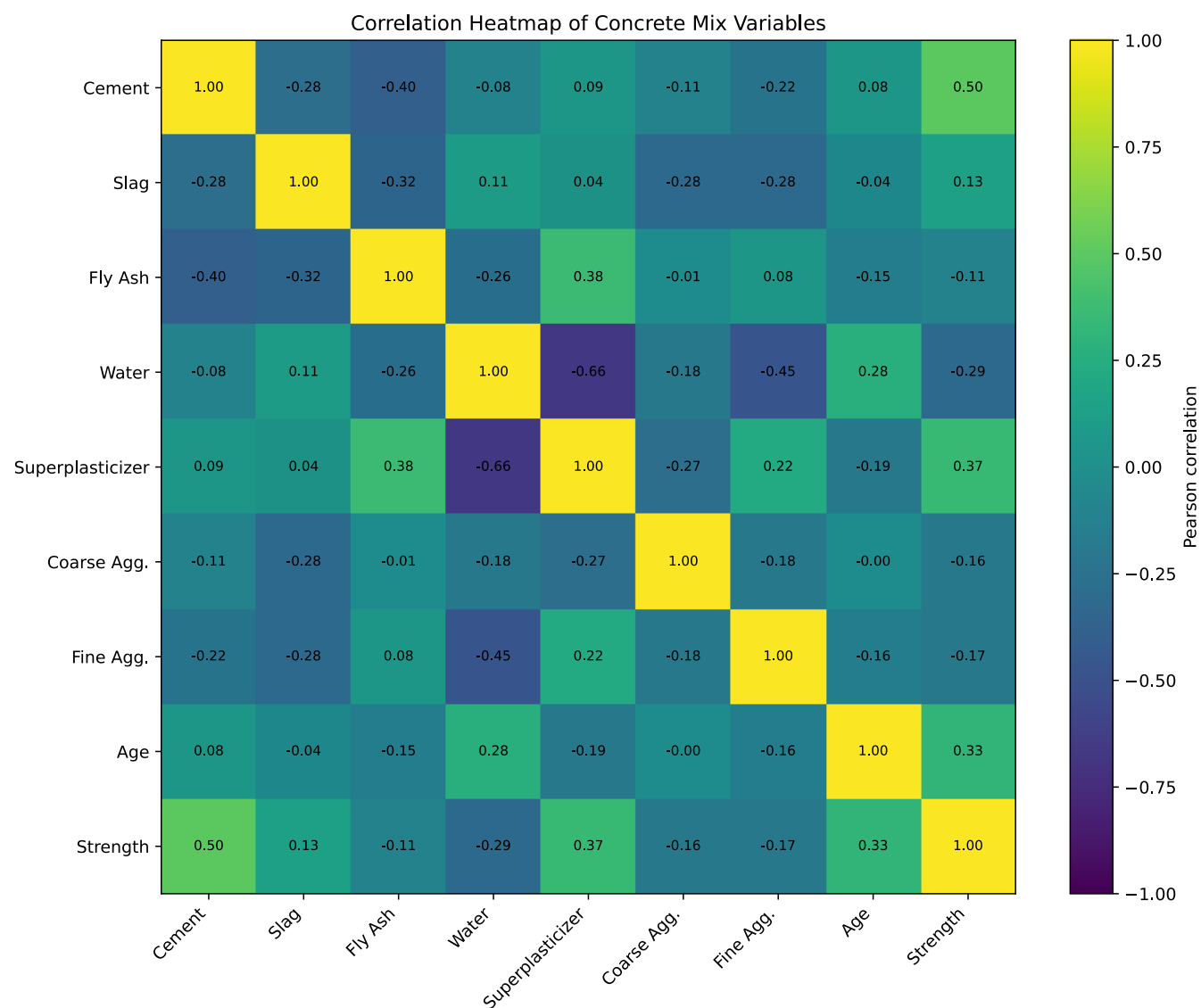


Figure 2. Correlation heatmap for the raw concrete mixture variables and compressive strength. Cement had the strongest positive raw correlation with strength, while water had the strongest negative raw correlation.

4. Data Preprocessing

The preprocessing workflow prepared the raw dataset for modeling while preserving as much valid data as possible. The team first checked the target column for missing values. There were 0 missing target rows, so no records were removed for missing target values. The team then removed duplicate rows before splitting the dataset. The executed preprocessing output found 25 duplicate rows and removed them, reducing the dataset from 1,030 rows to 1,005 rows.

The cleaned dataset was split into training and test sets using an 80/20 split with a fixed random state for reproducibility. This produced 804 training rows and 201 test rows. Because the task was regression rather than classification, no stratification was used. Missing-value checks after the split showed 0 missing values in the training predictors, test predictors, training target, and test target.

Outliers were handled using the IQR method on the training data only. The same lower and upper bounds were then applied to both training and test predictors by capping values outside the bounds. This choice reduced the influence of extreme values without discarding rows. It was especially important for curing age, where 6.09% of training values were outside the IQR bounds and long curing periods could exert substantial leverage. Capping was preferred over row removal because the dataset was modest in size and the extreme values still represented plausible concrete mixtures or curing conditions. Removing these rows would have reduced the training data available to the models and could have narrowed the range of mix designs represented in the final dataset. By deriving the capping bounds from the training data only, the preprocessing step reduced outlier influence without allowing information from the test set to shape the cleaning rules.

Table 1. Preprocessing decisions.

Column or Step	% Affected	Decision	Reasoning
Target strength	0.00%	Keep all rows	No target values were missing, so no target-based row removal was required.
Duplicate rows	2.43% of raw rows	Remove duplicates	The executed preprocessing output found 25 duplicate rows; removing them prevented repeated records from influencing model training and evaluation.
Cement	0.00% outliers	Keep	No training values fell outside the IQR bounds.
Blast furnace slag	0.12% outliers	Cap	A very small number of extreme slag values were capped to reduce leverage while retaining observations.
Fly ash	0.00% outliers	Keep	No training values were flagged by the IQR rule.
Water	1.74% outliers	Cap	Low or high water contents were capped because water was negatively related to strength and could influence ratio-based features.
Superplasticizer	1.12% outliers	Cap	Rare high admixture values were capped so they did not dominate the regression fit.
Coarse aggregate	0.00% outliers	Keep	Values fell within the IQR bounds.
Fine aggregate	0.62% outliers	Cap	A small number of extreme fine-aggregate values were capped.

Column or Step	% Affected	Decision	Reasoning
Age	6.09% outliers	Cap	Long curing periods up to 365 days were capped to reduce leverage while keeping all records.

The final cleaned modeling dataset contained 1,005 observations. The final split contained 804 training observations and 201 test observations. No imputation was required, and no rows were removed during outlier handling because capping was used instead of deletion.

5. Feature Engineering

Feature engineering expanded the original 8 predictors into a broader set of domain-informed variables. The team created total mixture quantities, ratio features, nonlinear transformations, and interaction terms. This process produced 40 total features, including the original 8 predictors and 32 engineered features. Because the expanded feature set included highly redundant variables, the team conservatively pruned 11 clearly overlapping engineered columns and retained 29 final modeling features.

No categorical encoding was required because all predictors were numeric. All 29 retained predictors were available in scaled form for Linear Regression, Ridge Regression, and Lasso Regression because those models are sensitive to feature scale. The target variable was excluded from scaling so that model error could still be interpreted directly in MPa. Tree-based models were trained using the unscaled retained feature set because Decision Tree, Random Forest, and Gradient Boosting models split features by thresholds and do not require standardized inputs. Since this was a regression problem rather than a classification problem, SMOTE was not applicable.

The feature engineering validation showed that engineered features substantially improved predictive performance in five-fold Ridge-style cross-validation. The baseline StandardScaler feature set achieved $CV R^2 = 0.7371$ and $CV RMSE = 8.3619$ MPa. The engineered StandardScaler feature set achieved $CV R^2 = 0.8474$ and $CV RMSE = 6.3622$ MPa. This improvement supported the decision to use engineered features in the modeling notebook.

The strongest engineered features also made domain sense. Concrete strength is not driven only by isolated ingredient amounts; it depends on how cementitious material, water, admixtures, aggregates, and curing time work together. Interaction features such as binder-age and cement-age helped represent strength development over time, while ratio features such as water-to-binder ratio helped represent mix balance. This was why the engineered feature set improved cross-validation performance compared with the raw scaled feature set.

Table 2. Engineered features.

Engineered Feature	Technique	Retained?	Rationale
total_binder	Total	Yes	Combined cement, slag, and fly ash to represent total cementitious material.
scm_total	Total	No	Summarized supplementary cementitious materials, but was pruned for redundancy.
aggregate_total	Total	No	Combined coarse and fine aggregate, but was pruned for redundancy.
paste_total	Total	No	Represented paste-like components, but was highly correlated with other binder measures.
water_cement_ratio	Ratio	Yes	Captured water relative to cement, a key concrete strength relationship.
water_binder_ratio	Ratio	Yes	Captured water relative to total binder; later became the second most important model feature.
binder_water_ratio	Ratio	No	Inverse of water-to-binder relationship; pruned due to overlap.
binder_minus_water	Difference	Yes	Measured binder surplus relative to water; had strong correlation with strength, $r = 0.6357$.
cement_binder_ratio	Ratio	Yes	Measured cement share of total binder.
slag_binder_ratio	Ratio	Yes	Measured slag share of total binder.
flyash_binder_ratio	Ratio	Yes	Measured fly ash share of total binder.
scm_binder_ratio	Ratio	No	Measured non-cement binder share, but was redundant with cement binder share.
sp_cement_ratio	Ratio	Yes	Measured admixture relative to cement.
sp_binder_ratio	Ratio	Yes	Measured admixture relative to total binder.
sp_water_ratio	Ratio	Yes	Measured admixture relative to water.
fine_coarse_ratio	Ratio	Yes	Captured aggregate balance between fine and coarse material.
coarse_fine_ratio	Ratio	No	Inverse of fine-to-coarse relationship; pruned for redundancy.
aggregate_binder_ratio	Ratio	Yes	Measured aggregate amount relative to binder.
paste_aggregate_ratio	Ratio	Yes	Measured paste-like components relative to total aggregate.
fine_total_agg_ratio	Ratio	Yes	Measured fine aggregate share of total aggregate.
coarse_total_agg_ratio	Ratio	No	Redundant with fine aggregate share and pruned.
log_age	Nonlinear transform	Yes	Captured diminishing strength gain over curing time.
sqrt_age	Nonlinear transform	No	Alternative nonlinear age transform; pruned for redundancy with age and log age.
age_squared	Nonlinear transform	Yes	Captured curvature in curing-age effects.
log_cement	Nonlinear transform	No	Captured diminishing cement effects but was highly correlated with cement.
log_water	Nonlinear transform	No	Captured nonlinear water effects but was highly correlated with water.
log_superplasticizer	Nonlinear transform	Yes	Captured nonlinear admixture effects and handled many low or zero values.
log_total_binder	Nonlinear transform	No	Captured nonlinear binder effects but was highly correlated with total binder.
cement_x_age	Interaction	Yes	Combined cement amount with curing age; had strong correlation with strength, $r = 0.7311$.
water_x_age	Interaction	Yes	Captured how water effects varied with curing age.
binder_x_age	Interaction	Yes	Combined total binder with curing age; strongest engineered correlation with strength, $r = 0.8212$.
sp_x_age	Interaction	Yes	Captured how superplasticizer effects varied with curing age.

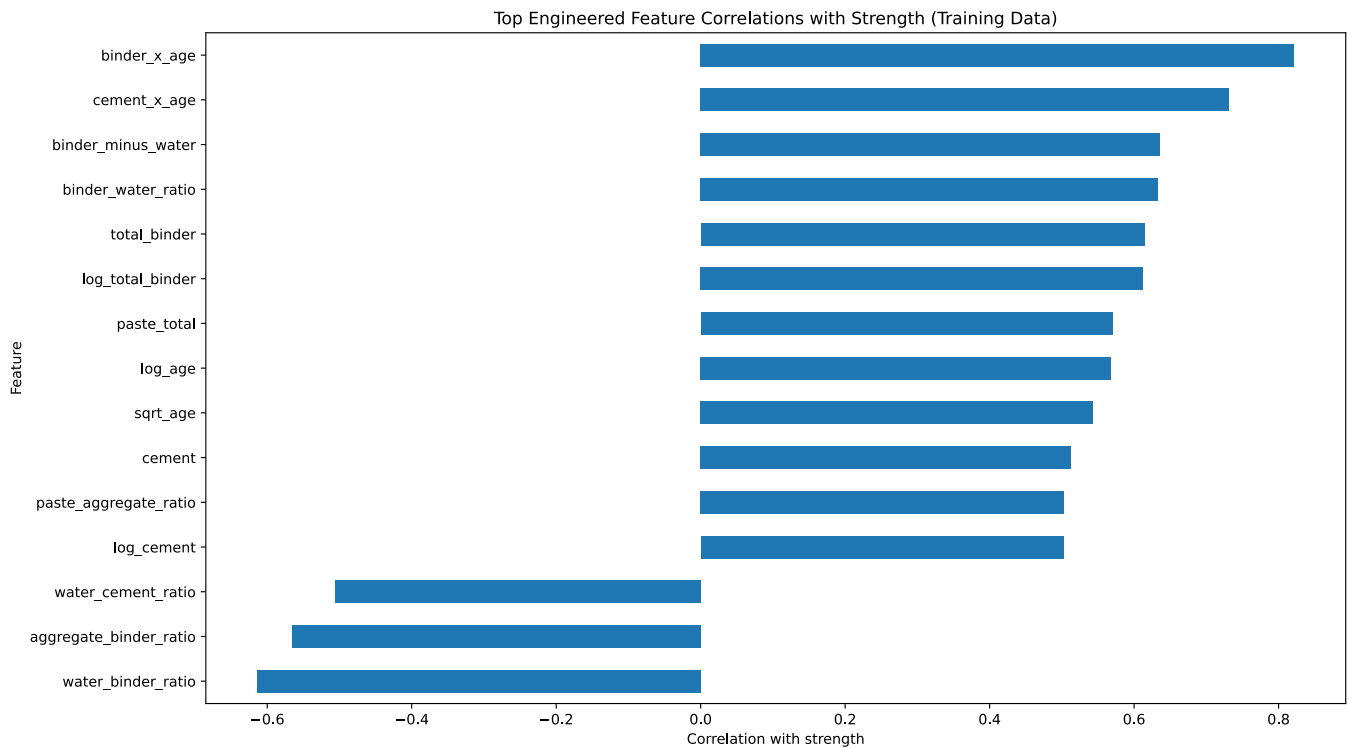


Figure 3. Top engineered feature correlations with compressive strength. Binder-age interaction, cement-age interaction, and binder-minus-water were the three strongest engineered relationships.

6. Modeling Approach and Results

The team trained a mix of interpretable linear models and more flexible tree-based models. Linear Regression, Ridge Regression, and Lasso Regression provided transparent baselines and helped evaluate whether the engineered features captured mostly linear relationships. Decision Tree, Random Forest, and Gradient Boosting models were included because concrete strength relationships are likely nonlinear and interaction-heavy. Hyperparameter tuning was then applied to Ridge Regression, Random Forest, and Gradient Boosting. Five-fold cross-validation was used to evaluate reliability across training folds before final test-set comparison.

The naive baseline predicted the training mean for every observation and had RMSE = 15.71 MPa. This baseline served as a simple reference point for how much the final model reduced prediction error. The best default model was Gradient Boosting, with test $R^2 = 0.9201$ and test RMSE = 4.4399 MPa. The final tuned Gradient Boosting model improved test R^2 to 0.9331 and reduced RMSE to 4.0616 MPa. It used a learning rate of 0.1, maximum tree depth of 3, and 300 estimators.

Table 3. Model results for default and tuned models.

Model	Version	R ² Train	R ² Test	RMSE Train	RMSE Test	CV R ² Mean	CV R ² Std
Gradient Boosting	Tuned	0.9873	0.9331	1.8497	4.0616	0.9230	0.0192
Random Forest	Tuned	0.9855	0.9226	1.9782	4.3690	0.9051	0.0138
Gradient Boosting	Default	0.9627	0.9201	3.1691	4.4399	0.9112	0.0187
Random Forest	Default	0.9843	0.9128	2.0567	4.6392	0.8946	0.0134
Linear Regression	Unscaled Default	0.8612	0.8426	6.1150	6.2306	0.8493	0.0180
Linear Regression	Scaled Default	0.8616	0.8425	6.1066	6.2341	0.8492	0.0185
Ridge Regression	Default	0.8595	0.8379	6.1527	6.3244	0.8499	0.0174
Ridge Regression	Tuned	0.8595	0.8379	6.1527	6.3244	0.8499	0.0174
Lasso Regression	Default	0.8584	0.8361	6.1773	6.3584	0.8492	0.0180
Decision Tree	Default	0.9879	0.7872	1.8033	7.2455	0.8140	0.0149

No model had cross-validation standard deviation above 0.03 in the final modeling comparison. This suggested that the final model rankings were reasonably stable across folds. However, the Decision Tree showed clear overfitting in the train-test results: its training R^2 was 0.9879, while its test R^2 was only 0.7872. The ensemble tree methods generalized better. Random Forest and Gradient Boosting both achieved test R^2 above 0.91, and tuned Gradient Boosting produced the strongest result.

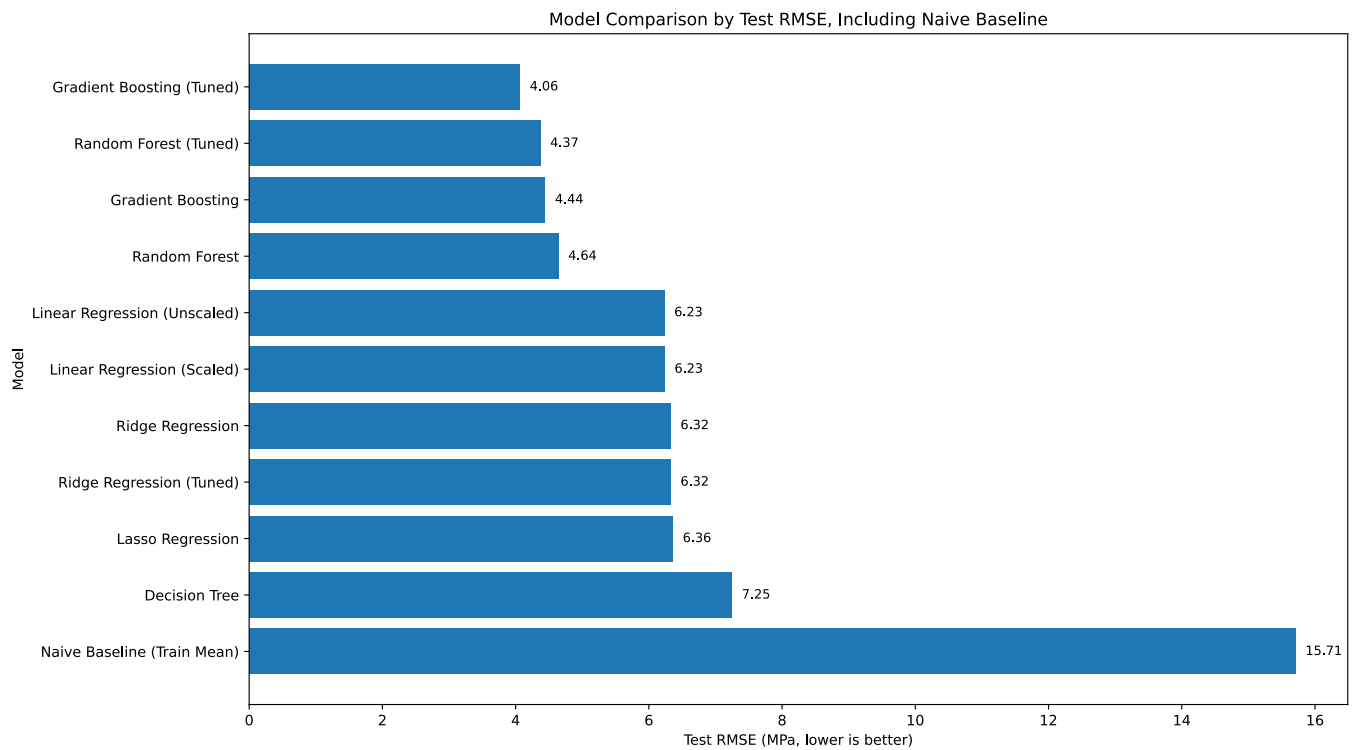


Figure 4. Model comparison by RMSE with the naive baseline. Tuned Gradient Boosting produced the lowest test RMSE at 4.0616 MPa, compared with the naive baseline RMSE of 15.71 MPa.

7. Model Selection and Interpretation

The final selected model was tuned Gradient Boosting. It had the best test-set performance of all models, with test $R^2 = 0.9331$ and test RMSE = 4.0616 MPa. It also had the strongest cross-validation result, with mean CV $R^2 = 0.9230$ and CV standard deviation = 0.0192. This cross-validation result supported the conclusion that the model's test performance was not caused only by a favorable train-test split.

The train-test gap for the final model was approximately 0.0542 R^2 points, based on training $R^2 = 0.9873$ and test $R^2 = 0.9331$. This gap showed some expected complexity from a powerful ensemble model, but the test performance remained strong and the cross-validation standard deviation remained below 0.03. Compared with the best default model, default Gradient Boosting, the tuned model improved test R^2 from 0.9201 to 0.9331. That was an absolute gain of approximately 0.0130 R^2 points. It also reduced RMSE from 4.4399 MPa to 4.0616 MPa, an improvement of approximately 0.3783 MPa.

Gradient Boosting was selected over Random Forest because it produced the strongest test performance and the strongest cross-validation mean while keeping cross-validation variability low. Tuned Random Forest also performed well, with test $R^2 = 0.9226$ and RMSE = 4.3690 MPa, but tuned Gradient Boosting improved test R^2 to 0.9331 and reduced RMSE to 4.0616 MPa. Because both ensemble models were stable across folds, the final selection was based on the better held-out test error and higher cross-validation mean.

The residual analysis supported the model selection. The actual-versus-predicted pattern mostly followed the diagonal, showing strong overall agreement between predicted and observed compressive strength. The mean residual was about 0.13 MPa, which suggested little overall bias. However, a few high-error cases remained, especially among moderate-to-high predicted strengths. This indicated that the model was useful for screening candidate mixes but should not replace laboratory validation for safety-critical decisions.

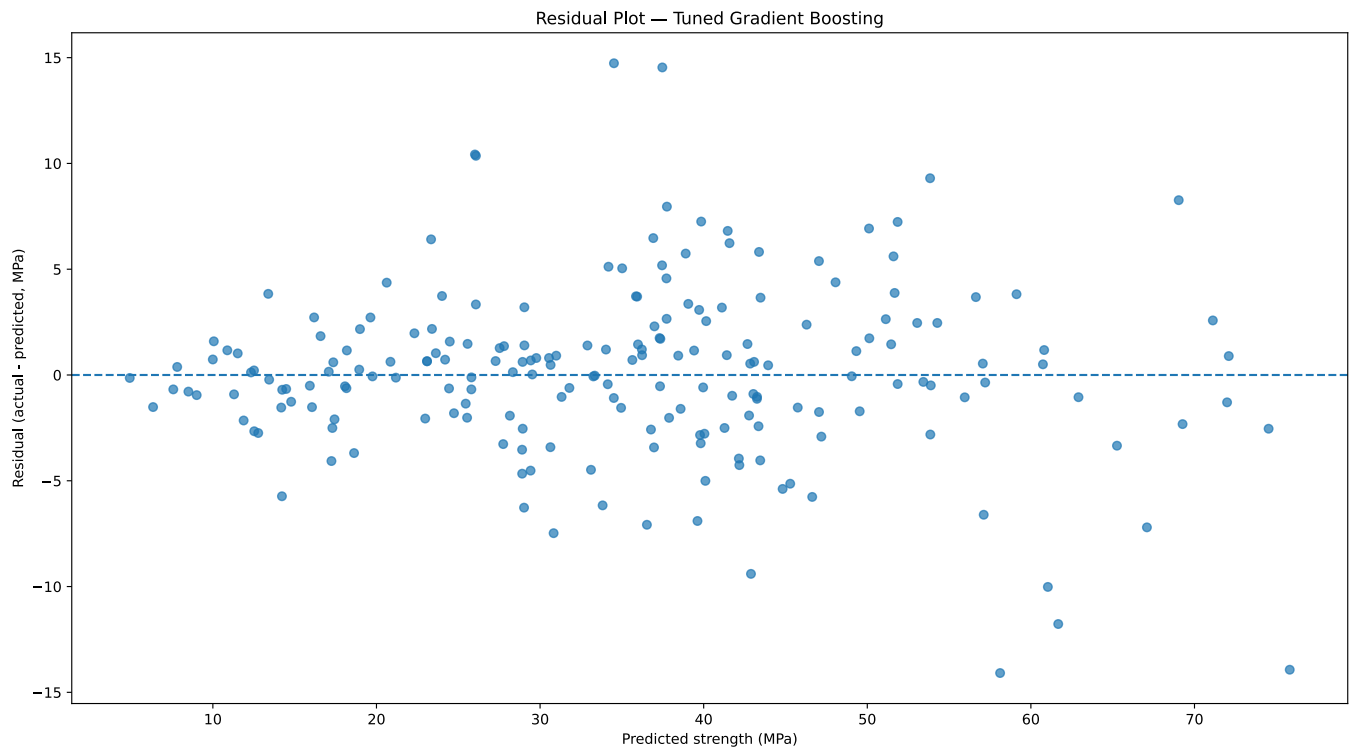


Figure 5. Residual plot for tuned Gradient Boosting. Residuals were centered near zero, with mean residual about 0.13 MPa, but a few higher-error predictions remained.

The final model's top three features were binder-age interaction, water-to-binder ratio, and cement-age interaction. Binder-age interaction had importance 0.5234, making it the dominant predictor. It measured the combined effect of total cementitious binder and curing time. This was meaningful because concrete strength development depends on both available cementitious material and time for hydration. The business implication was that candidate mix screening should evaluate binder content together with curing expectations, not in isolation.

Water-to-binder ratio was the second most important feature, with importance 0.1663. It measured the amount of water relative to total binder. This aligned with concrete-domain intuition because excess water can reduce compressive strength by increasing porosity after curing. The business implication was that water-to-binder balance should be monitored carefully when comparing otherwise similar mixes.

Cement-age interaction was the third most important feature, with importance 0.1472. It measured how cement content and curing time worked together. This feature reinforced the conclusion that the model learned strength development patterns rather than only static ingredient effects. The business implication was that predictions should be interpreted in the context of curing age; a mix that appears underperforming at an early age may perform differently after longer curing.

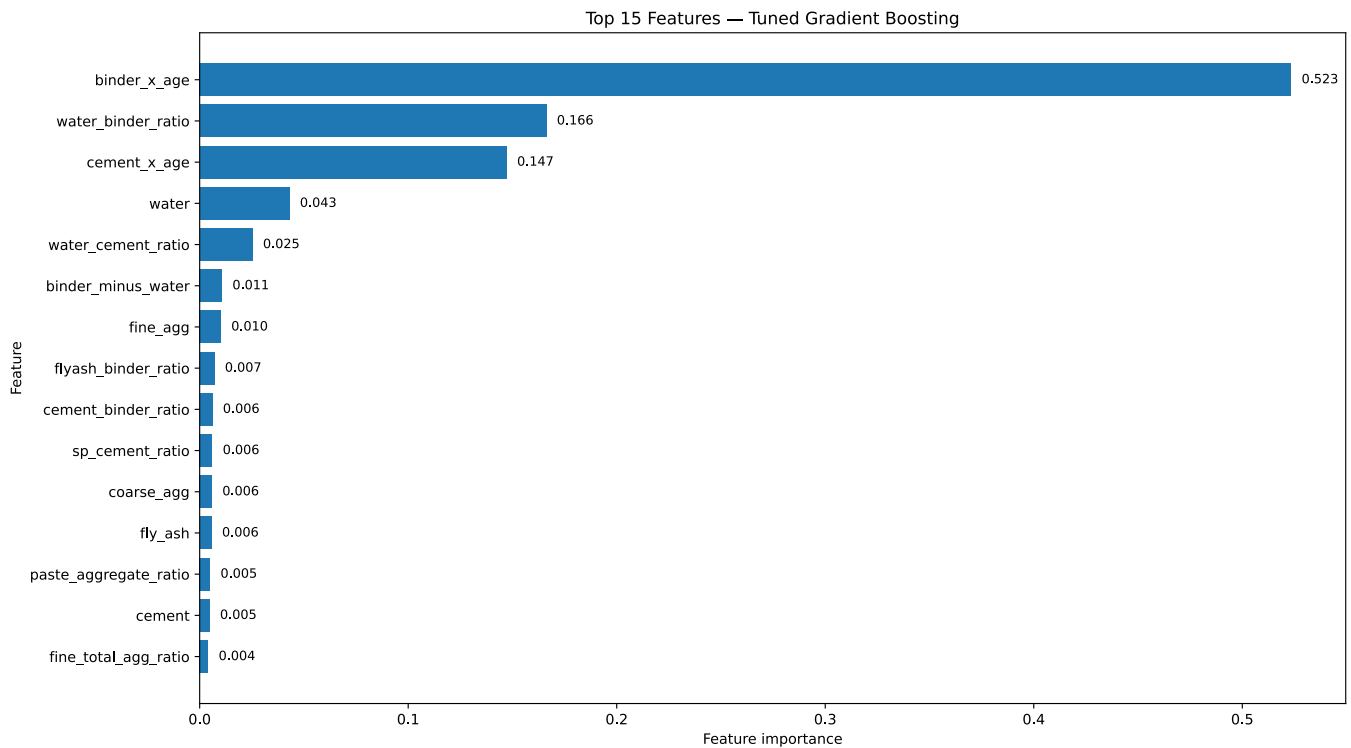


Figure 6. Feature importance for the tuned Gradient Boosting model. Binder-age interaction, water-to-binder ratio, and cement-age interaction were the top three predictors.

8. Conclusions and Recommendations

This project predicted concrete compressive strength from mixture components and curing age. The best model was tuned Gradient Boosting, which achieved test $R^2 = 0.9331$ and test RMSE = 4.0616 MPa, substantially improving on the naive baseline RMSE of 15.71 MPa.

The first recommendation was to use the final model as an early screening tool for candidate concrete mixes, not as a replacement for physical testing. The model reduced RMSE by about 11.64 MPa compared with the naive baseline, which made it useful for narrowing candidate designs before lab validation. However, because some high-error cases remained, final decisions should still rely on validated compressive strength testing.

The second recommendation was to prioritize binder-age balance when evaluating candidate mixes. The binder-age interaction was the strongest feature in the final model, with importance 0.5234, and it also had the strongest engineered correlation with strength, $r = 0.8212$. This suggested that mix design decisions should consider cementitious binder content together with curing age rather than treating age as a secondary input.

The third recommendation was to monitor water-to-binder ratio closely during mix screening. Water-to-binder ratio was the second most important feature in the final model, with importance 0.1663, and had a strong negative engineered correlation with strength of approximately $r = -0.6133$. In practical terms, candidate mixes with high water relative to binder

should be reviewed carefully because they may be more likely to fall below expected compressive strength.

The first limitation was dataset size and source. The cleaned dataset contained 1,005 rows from a public dataset rather than local production data. The model may therefore perform differently on local materials, suppliers, batching processes, or regional construction conditions. The second limitation was missing real-world variables. The dataset did not include curing temperature, humidity, cement type, admixture chemistry, material cost, environmental impact, slump, air content, or site conditions. These missing variables could explain some of the remaining high-error predictions.

The next step should be to validate the tuned Gradient Boosting model on local or project-specific concrete batch records. A useful validation set should include the same mixture variables and curing age, but should also add cost and environmental variables where available. This would allow the model to support not only strength screening but also practical mix optimization for cost and sustainability.

9. References

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